M.TECH AIML • COMPUTER VISION

Unit-I: Mathematical Morphology

# Medical Image Morphological Processing Using Brain MRI

Non-learning OpenCV pipeline for structural segmentation, binarization, and boundary extraction.

**Dataset Source**
Brain MRI Scan (Neg0.jpg)

**Core Stack**
Python • OpenCV • NumPy

PROJECT SCOPE

# Mathematical Morphology Objectives

### Light Normalization

Standardized 256×256 grayscale matrix.

### Otsu Binarization

Automated tissue-background segmentation.

### Set Morphology

Erosion, Dilation, Opening, Closing.

### Boundary Search

Internal contour via set difference.

STEP 01 PIPELINE

# Brain MRI Acquisition & Normalization

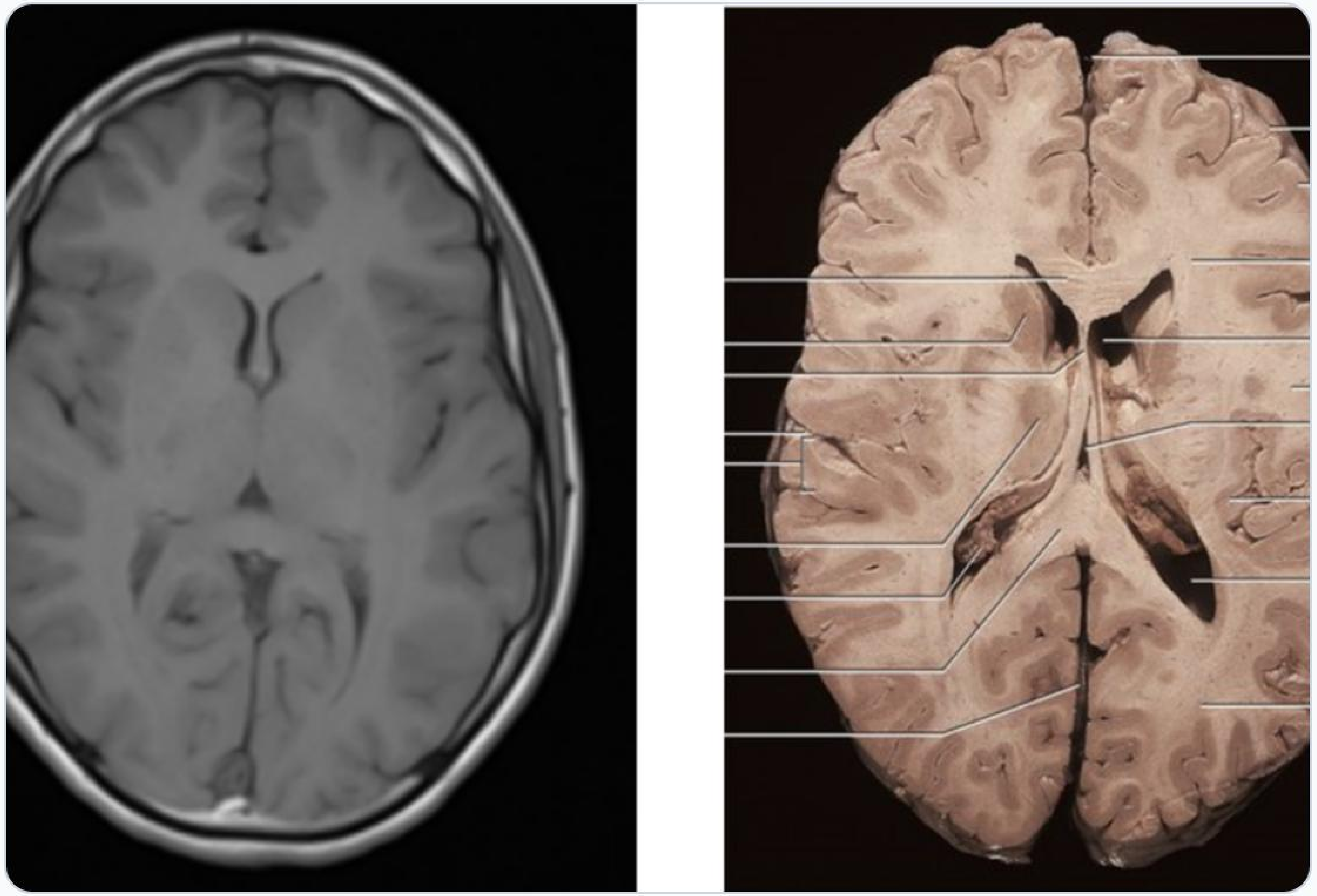
**Automated Multi-Path Loader:** Searches local repository and mounted Google Drive dynamically.

**Single-Channel Conversion:** Loads scan directly into 8-bit grayscale space (`cv2.IMREAD_GRAYSCALE`).

**Spatial Resampling:** Standardizes input to 256×256 matrix resolution.

**Deterministic Pipeline:** Eliminates machine learning training dependencies.

> **Output Dimension: 256 × 256 Pixel Matrix**



STEP 02 PIPELINE

# Intensity Distribution & Contrast Metrics

MEAN INTENSITY (μ)

**119.79**

Average global brightness

STD DEVIATION (σ)

**49.75**

Intensity variance & contrast

DYNAMIC RANGE

**[12, 255]**

Full 8-bit scale utilization

### Bi-Modal Intensity Profile (Background vs Tissue)

STEP 03 PIPELINE

# Otsu Automatic Binarization

OPTIMAL THRESHOLD (T*)

**95**

Calculated via intra-class variance minimization.

### Non-Parametric Segmentation

No manual intensity tuning required.

### Binary Matrix Generation

Converts image to boolean {0, 255} state.

### Tissue Separation

Isolates brain parenchyma from background air.

STEP 05 PIPELINE

# Morphological Boundary Extraction

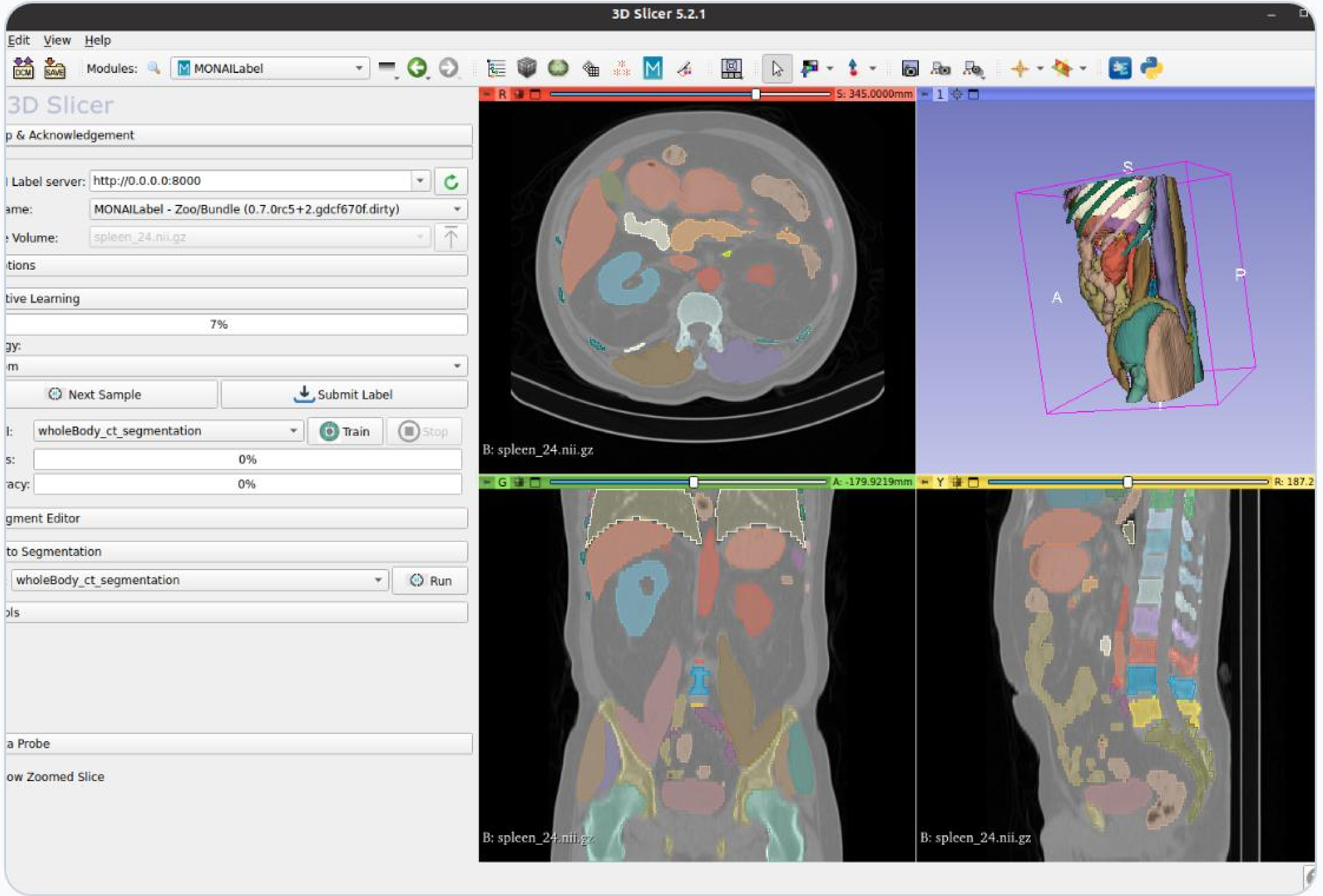
MATHEMATICAL FORMULA

$$\beta(A) = A - (A \ominus B)$$

**Set Difference:** Subtracts eroded mask from original binary mask.

**Single-Pixel Contour:** Produces exact 1-pixel structural outline.

**Red Channel Overlay:** Superimposes boundary onto grayscale scan.



# Structuring Element Scale Impact

3 × 3 Kernel

### Mild Thinning

Preserves fine anatomical details.

5 × 5 Kernel

### Moderate Shrinking

Smoothes structural outer edges.

7 × 7 Kernel

### Severe Loss

Erodes critical thin structures.

STEP 07 PIPELINE

# Master Pipeline Execution Flow

1. **Input & Normalization**
   Grayscale read and $256 \times 256$ matrix resize.

2. **Otsu Thresholding**
   Auto-bimodal cutoff calculation ($T^* = 95$).

3. **Morphological Filter Suite**
   Apply Erosion, Dilation, Opening, and Closing.

4. **Boundary & Overlay Rendering**
   Extract contour difference and render red overlay.

IMPACT

# Clinical & Computational Takeaways

### Ultra-Fast Execution

Sub-millisecond processing per frame.

### Pre-Segmentation Utility

Ideal preprocessing for deep learning.

### Deterministic Precision

Zero training dataset requirements.